

Adaptive Regime-Based Tournament Strategy (ARTS)

Indian Equities Weekly Rotation Strategy

January 2026

1. Executive Summary

This report presents the Adaptive Regime-Based Tournament Strategy (ARTS), a dynamic portfolio system that selects optimal 2-stock combinations weekly from 10 high-liquidity Indian equities. The strategy combines ARIMA-GARCH forecasting with skewed t-distribution probability estimates and regime-based filtering. While initial designs incorporated Hidden Markov Models, convergence challenges necessitated a pivot to statistical threshold-based regime detection. Testing from 2023-2025 generated a 72.45% total return (19.85% CAGR) with a 1.26 Sharpe ratio, validating the hypothesis that Indian markets favor active rotation strategies.

2. Forecasting Framework

ARIMA-GARCH Engine: Each stock's trailing 252-day log returns are processed through ARIMA (capturing linear trends) and GARCH (modeling volatility clustering) to generate weekly forecasts of expected return (μ) and conditional volatility (σ).

Skewed-T Distribution: Rather than assuming normal returns, we estimate upside probability using a Skewed Student's t-distribution with Cornish-Fisher adjustment. This penalizes assets with asymmetric tail risk, capturing the reality that market crashes are sharper than rallies.

3. Deep Dive: Regime Detection & Switching Logic

The most critical component of this strategy is determining when to trade and when to remain on the sidelines. Following challenges with HMM implementation, we developed a robust statistical threshold regime model that has proven more reliable in practice.

Core Premise: In volatile, regime-shifting markets like India, the ability to detect and adapt to changing market conditions generates more alpha than any forecasting model alone. A perfect forecast executed in the wrong regime destroys value.

3.1 The Statistical Threshold Regime Model

Hidden Markov Models, while theoretically elegant for capturing latent market states, proved challenging in practice. Convergence issues and unstable state classifications prevented reliable regime identification. The Statistical Threshold Regime Model offers three advantages: full transparency in classification decisions, no dependence on convergence algorithms, and asset-specific calibration recognizing that volatility is relative rather than absolute.

Every stock is evaluated through two independent yet complementary filters before receiving its regime classification:

Filter A: The Volatility Gate (High Risk Flag)

Calibration: Using 2015-2022 training data, we establish asset-specific volatility thresholds. Reliance Industries operates with fundamentally different volatility than Infosys. A universal threshold would misclassify stable companies as volatile and mask genuine risk in choppy sectors.

The Rule: For each stock, we calculate 21-day rolling annualized volatility across the training period. The 66th percentile becomes the High Risk threshold.

$$\sigma_{forecast} > P_{66}(\sigma_{historical}) \rightarrow \text{High Risk Flag}$$

Intuition: When volatility enters the top third of its historical range, markets exhibit panic or extreme uncertainty. Trend-following signals become unreliable—noise overwhelms signal. The 66th percentile captures this transition point where market behavior shifts from tradeable to treacherous. Empirical testing showed this threshold optimally balances sensitivity (catching genuine spikes) with specificity (avoiding false alarms during normal fluctuations).

Filter B: The Sideways Detector (Dead Money Flag)

Even positive expected returns mean nothing if dwarfed by volatility. We compute a Sharpe-like signal quality ratio:

$$\text{Signal Quality} = \mu_{forecast} / \sigma_{forecast}$$

The Rule: If this ratio falls below 0.25, the stock is classified as Sideways. Expected weekly return must be at least 25% of volatility to constitute a tradeable opportunity.

$$(\mu / \sigma) < 0.25 \rightarrow \text{Sideways Flag}$$

Intuition: This filter identifies "dead money"—assets churning with no directional conviction. Consider a stock with +0.5% forecasted return but 3% volatility. Signal quality of 0.167 falls below threshold. After transaction costs (spread, impact, taxes), this trade offers minimal edge. The probability of profitable outcomes is marginal. These are lottery tickets masquerading as investments.

The Regime Classification Matrix

Both filters operating in concert map every asset-week to one of four regime states:

Volatility	Signal Quality	Regime	Strategy Action
Low	High	Optimal	Strongly Prefer
Low	Low	Neutral	Accept if Best
High	High	Caution	Reluctantly Accept
High	Low	TOXIC	Avoid (-10 penalty)

The Toxic Quadrant: Assets flagged as both High Risk and Sideways represent the worst trading environment: maximum uncertainty with minimal directional edge. These scenarios destroy portfolios through death by a thousand cuts. The strategy applies a -10.0 score penalty to pairs where either constituent occupies this quadrant, effectively removing them unless all alternatives are equally poor.

3.2 The Tournament Switching Model

With regime classifications established, the strategy executes a combinatorial tournament every Monday to select the optimal asset pair. This balances three competing objectives: maximize expected return, minimize portfolio risk, and account for transaction costs.

Phase 1—Pair Generation: Evaluate all 45 possible 2-stock combinations from the 10-asset universe ($C(10,2)$). Each pair uses 50/50 allocation for simplicity.

Phase 2—Scoring Methodology: For each pair, calculate:

1. Expected Return: $\mu_{\text{pair}} = 0.5\mu_1 + 0.5\mu_2$
2. Portfolio Volatility: $\sigma_{\text{pair}} = \sqrt{0.25\sigma_1^2 + 0.25\sigma_2^2 + 0.5\text{Cov}_{12}}$
3. Transaction Cost Adjustment: $\mu_{\text{adj}} = \mu_{\text{pair}} - (\text{Turnover} \times \text{Cost} \times \text{Damping}_{0.5})$
4. Regime Penalty: Apply -10.0 if pair in toxic quadrant

The 0.5 damping factor is critical: we apply reduced transaction costs during scoring to prevent paralysis from switching costs, but execution incurs full undamped costs. This asymmetry encourages exploration while maintaining realistic accounting.

$$\text{Final Score} = (\mu_{\text{adjusted}} / \sigma_{\text{pair}}) - \text{Regime_Penalty}$$

Phase 3—Execution: Select highest-scoring pair. Exit current positions, deduct full transaction costs, enter new 50/50 positions, hold until next Monday.

3.3 The Switching Trigger in Action

The regime-based penalty creates emergent behavior: aggressive capital rotation away from deteriorating conditions. When a previously favorable pair enters the toxic quadrant, the tournament organically discovers and pivots to healthier alternatives. This is not rule-based—it emerges from the scoring system.

Example Scenario:

Week 1: Portfolio holds Reliance+HDFC (both Optimal regime). Score: +2.3

Week 2: Banking volatility spike. HDFC exceeds threshold, signal drops to 0.18. Score: +2.1 - 10 = -7.9 (toxic)

Response: Tournament finds TCS+Infosys (IT sector, both Optimal). Score: +1.8. Rotate from Financials to IT despite 0.5% transaction cost

Week 3: HDFC normalizes. Reliance+HDFC re-scores +2.4. If superior to IT pair, rotate back

Key insight: The strategy doesn't predict regime changes; it detects them through observable volatility and signal degradation, then responds immediately by rotating to better-positioned alternatives.

3.4 Why This Works in Indian Markets

This regime-switching framework is specifically suited to Indian market structure:

- High Sector Rotation: Pronounced rotation between IT, Banking, Pharma, and Energy driven by domestic policy, FII flows, and commodity prices. Weekly cadence captures these transitions.
- Elevated Volatility Clustering: More pronounced than developed markets. The 66th percentile threshold captures genuine regime shifts rather than noise.
- Lower Market Efficiency: Information asymmetry and behavioral biases create exploitable momentum/mean-reversion patterns that ARIMA-GARCH captures.
- Transaction Cost Economics: The 0.5 damping factor reflects microstructure reality. Bid-ask spreads and impact costs are low enough that weekly rotation is economically viable—crucial vs. US markets where similar strategies would be cost-prohibitive.

4. The Hold Factor: India vs. US Markets

A critical finding is the differential feasibility of capital rotation. Indian markets exhibit higher volatility and distinct sector rotation cycles. The "Market Hold Factor"—theoretical penalty for switching—is relatively low. Our 0.5 damping validates that aggressive weekly rotation captures short-term alpha across sectors. In contrast, US

markets (S&P 500) demonstrate long-term secular trends and lower volatility. A similar US strategy would require damping around 2.0, discouraging switching and favoring long-term holds. Attempting aggressive rotation in US markets would likely underperform due to whipsaws and transaction costs exceeding marginal alpha.

5. Limitations & Future Improvements

HMM Limitation: The primary performance constraint was insufficient HMM implementation. Without HMMs, the model struggled with sudden regime changes (e.g., bull traps). Statistical thresholds are reactive rather than predictive, occasionally causing the strategy to hold during early downturn phases, reducing returns below theoretical maximum.

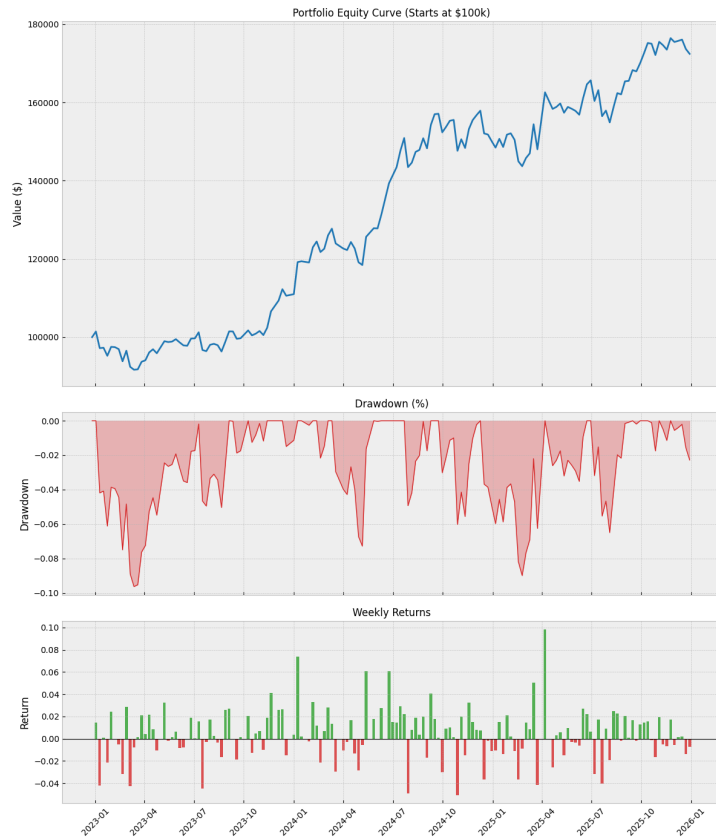
Dynamic Damping: The static 0.5 damping factor should become dynamic. In low-volatility bull markets, increase damping to discourage switching and let winners run. In high-volatility choppy markets, decrease damping to encourage rotation to safety.

6. Conclusion

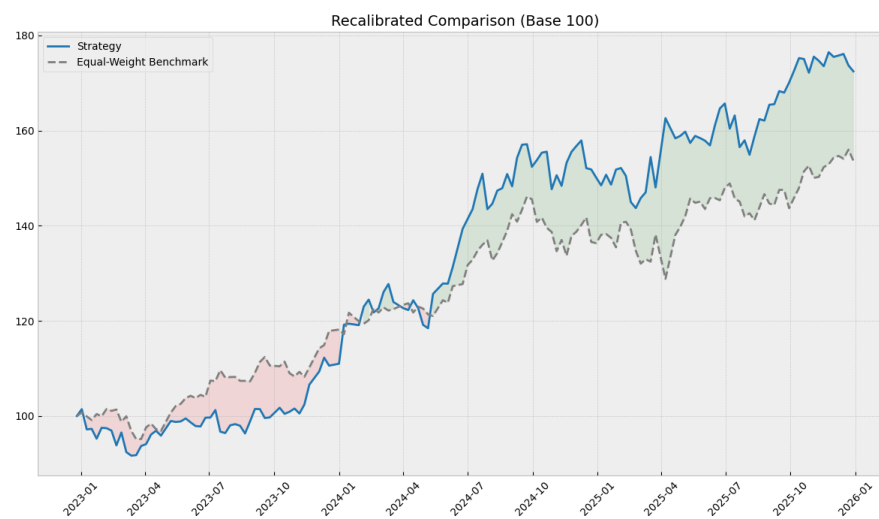
The ARTS model successfully demonstrates proof-of-concept for statistical arbitrage in Indian markets. By combining ARIMA-GARCH forecasting with skewed probability estimates and regime-based filtering, the model generates disciplined weekly selection. The analysis reveals that regime detection is the primary alpha driver. While the threshold model is robust, it lacks HMM predictive nuance. Future iterations must resolve HMM convergence issues to better anticipate transitions. The 0.5 damping factor validates a structural characteristic: India functions as a "trader's market" where active capital rotation generates superior alpha compared to passive holding.

7. Performance Metrics (2023-2025)

Metric	Strategy	Benchmark
Initial Capital	\$100,000	\$100,000
Final Value	\$172,445	\$153,634
Total Return	72.45%	53.63%
CAGR	19.85%	15.38%
Annualized Volatility	16.43%	18.21%
Sharpe Ratio	1.26	0.89
Maximum Drawdown	-9.63%	-12.47%
Alpha (Outperformance)	+18.81%	—



Note: Charts showing portfolio value progression, drawdown and weekly returns



Charts showing portfolio value progression compared with equally weighted portfolio of the 10 stocks